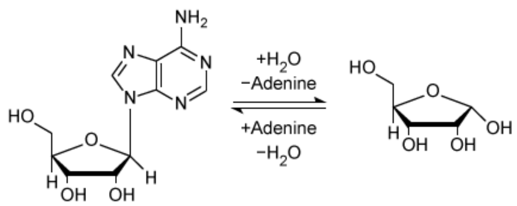
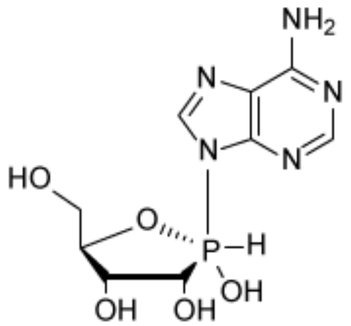
Enzyme X catalyzes the hydrolytic depurination of adenosine. The enzyme-catalyzed reaction proceeds via an S_N2 mechanism, where water acts as the incoming nucleophile.



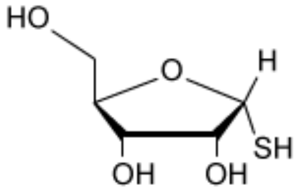
Based on this information, and given that enzymes bind most tightly to the transition state of their reaction, which of the following molecules most likely has the smallest K_d and serves as the best competitive inhibitor for Enzyme X?

A.



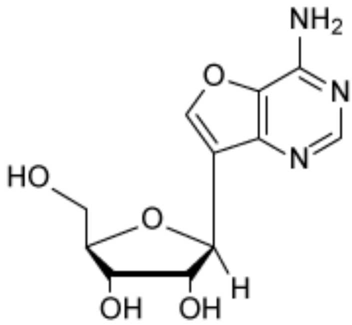
[38%]

B.



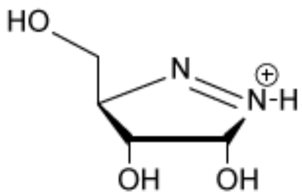
[11%]

C.



[32%]

D.

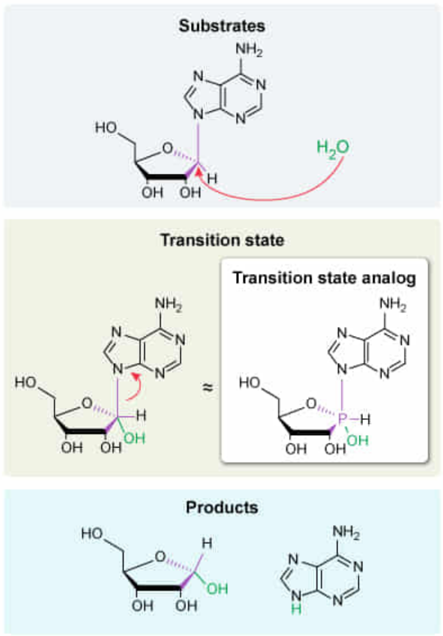


[16%]

Correct
39% answered correctly

Explanation:

S_N2 reaction catalyzed by enzyme X



Molecules that resemble the transition state of a reaction (transition state analogs) bind tightly and are good competitive inhibitions

Enzymes are biological catalysts that speed up chemical reactions, allowing them to occur under physiological conditions. One of the main factors that determines a reaction's rate is its activation energy, which is the energy difference between the reactants and the high-energy transition state (‡) of the reaction. Enzymes can decrease the activation energy barrier by **binding and stabilizing the transition state**. Consequently, **transition state analogs** (ie, molecules that **structurally resemble** the transition state) bind tightly to the enzyme, have **small** K_d values, and can serve as highly potent competitive inhibitors.

The question states that the enzyme-catalyzed reaction proceeds via an S_N2 mechanism. S_N2 reactions occur when an incoming nucleophile attacks a tetrahedral electrophile on the side opposite the leaving group (ie, backside attack). This reaction involves a transient trigonal bipyramidal transition state with the electrophile "bonded" to **five different atoms**: the incoming nucleophile (forming a bond), the outgoing leaving group (breaking a bond), and the three other atoms that are unaffected by the reaction. Once the incoming nucleophile–electrophile bond forms completely and the electrophile–leaving group bond fully breaks, the reaction is complete.

In the given reaction, adenine is the leaving group, water is the nucleophile, and the anomeric carbon of ribose is the electrophile. Replacing the anomeric carbon with a phosphorus atom allows five bonds to stably exist at that position (phosphorus can form sp^3d hybridized orbitals to have five bonds; carbon cannot). The resulting molecule, shown in Choice A, resembles the **trigonal bipyramidal transition state** of the reaction and is likely to bind the enzyme most tightly (lowest K_d) of the options given. Therefore this molecule is likely the best competitive inhibitor of Enzyme X.

(Choices B and C) These are structural analogs of one of the *products* (ribose) and one of the *reactants* (adenosine) of the reaction, respectively. Although these *can* act as competitive inhibitors, they are not expected to have the strongest binding (ie, lowest K_d) of the given choices.

(Choice D) This structure contains a trigonal planar cation at the analogous position to the anomeric carbon. This would be a good analog for an S_N1 intermediate but *not* an S_N2 transition state.

Educational objective:
Enzymes catalyze reactions by binding and stabilizing the transition state. Transition state analogs are molecules that structurally resemble a reaction's transition state, and therefore strongly bind and inhibit an enzyme.

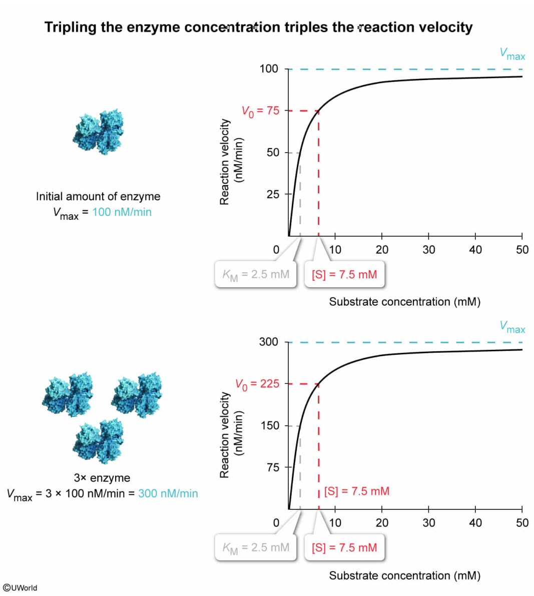
Time spent:0 sec	**QID**:403737
Subject:	**System:**
Biochemistry	Enzymes

The Michaelis–Menten kinetic parameters were measured for enzyme E with substrate S, yielding a K_M value of 2.5 mM and a V_{max} of 100 nM/min. If this first experiment were to be repeated with triple the concentration of enzyme, what would be the initial reaction velocity (V_0) for this second trial when [S] = 7.50 mM?

- A. 75 nM/min [41%]
- B. 100 nM/min [14%]
- C. 150 nM/min [19%]
- ✔ D. 225 nM/min [24%]

Correct
24% answered correctly

Explanation:



Enzyme kinetics—the study of the reaction rates of enzyme-catalyzed reactions—are frequently described using the Michaelis–Menten equation, which relates the initial reaction velocity (V_0) to the concentration of free substrate given certain assumptions. The Michaelis–Menten equation includes two parameters: the Michaelis constant (K_M) and the maximum reaction velocity (V_{max}).

The K_M parameter is an intensive property of the enzyme and does not vary with the amount of enzyme present. The V_{max} parameter is an **extensive property** and *does* **vary** with the **amount of enzyme** present. Mathematically, V_{max} is the product of the intensive property k_{cat} (ie, the number of reactions catalyzed per unit time for a single enzyme molecule under saturating substrate conditions) and the extensive property of enzyme concentration, making V_{max} **proportional** both to k_{cat} and to enzyme concentration.

The question states that the repeated experiment would contain **triple** the amount of enzyme as the first experiment. Because V_{max} is proportional to enzyme concentration, V_{max} will **triple** as well. The Michaelis–Menten equation can then be solved for this scenario, substituting the first experiment's V_{max} of 100 nM/min for the tripled value of 300 nM/min.

$$V_0 = V_{\text{max}}\left(\frac{[\text{S}]}{K_\text{M} + [\text{S}]}\right)$$

$$V_0 = 300\frac{\text{nM}}{\text{min}}\left(\frac{7.5\ \text{mM}}{2.5\ \text{mM} + 7.5\ \text{mM}}\right)$$

$$V_0 = 300\frac{\text{nM}}{\text{min}}\left(\frac{7.5\ \text{mM}}{2.5\ \text{mM} + 7.5\ \text{mM}}\right)$$

$$V_0 = 300\frac{\text{nM}}{\text{min}}\left(\frac{7.5\ \cancel{\text{mM}}}{10\ \cancel{\text{mM}}}\right)$$

$$V_0 = 300\frac{\text{nM}}{\text{min}}\left(\frac{3}{4}\right)$$

$$V_0 = 225\frac{\text{nM}}{\text{min}}$$

(Choice A) 75 nM/min is the V_0 for 7.5 mM of substrate using the *first experiment's* enzyme concentration. The repeated experiment will contain *triple* the enzyme, which will triple the V_0.

(Choice B) This is the first experiment's V_{max} value, which is V_0 when the enzyme is saturated with substrate. However, the given substrate concentration will cause only 75% saturation.

(Choice C) 150 nM/min is half the value of V_{max} for the tripled enzyme condition. This value would be achieved if [S] = K_M; however, [S] = 7.5 mM and K_M = 2.5 mM.

Educational objective:
In Michaelis–Menten experiments, maximum reaction velocity (V_{max}) is proportional to the concentration of enzyme used in that experiment. Tripling the amount of enzyme will triple the V_{max} value.

Time spent:0 sec
Subject:
Biochemistry

QID:403738
System:
Enzymes